

Housing Supply Elasticity and Transit Welfare: A Replication and Extension of Severen (2021)

Marko Ikävalko & Mikko Viikari

May 2026

Abstract

Severen (2021) estimates the general-equilibrium welfare effects of the Los Angeles Metro Rail network. The paper finds commuting flows rose 11–16 % between station-adjacent census-tract pairs and derives an aggregate welfare gain of roughly \$94 million per year (our replication: \$93.55 million), holding the city-wide inverse housing supply elasticity ψ at its pooled estimate of 1.602 (a supply elasticity of $1/\psi \approx 0.62$). We replicate the full Severen pipeline and extend the analysis in two directions. First, we attempt to estimate ψ separately for geographic sub-regions of the LA metro area (by coastal proximity, density quartile, and county group), motivated by the substantial spatial variation in local housing markets. The Bartik shift-share instrument proves too weak for sub-regional identification: first-stage F -statistics fall well below conventional thresholds in every partition tried (KP $F < 5$, often < 1), against the paper’s pooled KP $F \approx 12$. Second, unable to obtain reliable sub-regional estimates, we vary ψ over a wide range in the calibrated GE model. We find the aggregate welfare estimate is essentially invariant to ψ : across a $5\times$ range (0.75–4.0), welfare moves by only \$0.33 million (0.35 %). This is expected for a small intervention in a car-dominated city, where the transit-induced demand shock near stations is too modest for housing supply conditions to affect aggregate welfare — though, as we discuss, ψ still governs the *incidence* of the gains. “Amplified-connectivity” counterfactuals confirm that larger commuting benefits have limited impact: doubling the commuting disutility reduction associated with Metro connectivity adds only \$105 million per year in aggregate welfare.

Keywords: transit infrastructure, general equilibrium, housing supply elasticity, welfare analysis, Los Angeles

Contents

1	Introduction	2
2	The Severen (2021) Framework	3
2.1	Setting and data	3
2.2	Commuting flow estimates	3
2.3	Structural elasticities	3
2.4	General-equilibrium welfare model	4
3	Modifications and Methods	4
3.1	Motivation: heterogeneous housing supply	4
3.2	Attempted sub-regional estimation	5
3.3	Identification failure: instrument weakness	5
3.4	GE sensitivity analysis	6
3.5	“Amplified-connectivity” counterfactuals	6
4	Results	7
4.1	Aggregate welfare is insensitive to ψ	7
4.2	Sub-regional ψ estimates: instrument weakness	8
4.3	“Amplified-connectivity” counterfactuals	9
4.4	The mechanism: why ψ does not move aggregate welfare	10
5	Conclusion	11

1 Introduction

Urban transit investment is among the most capital-intensive interventions available to city governments. Evaluating whether such investments are welfare-improving requires a framework that accounts not only for the direct benefits to commuters but also for the general-equilibrium adjustments in labour and housing markets that follow. [Severen \(2021\)](#) provides one of the most complete such evaluations to date, applying a quantitative spatial model to the 1990–2002 build-out of the Los Angeles Metro Rail network.

The paper treats housing supply conditions as a single city-wide constant: the inverse housing supply elasticity ψ , estimated at 1.602 using a Bartik shift-share instrument (implying a housing supply elasticity of $1/\psi = 0.624$, i.e., inelastic supply). This is a natural baseline, but it is a strong assumption: housing markets in the five-county LA metro area vary enormously across coastal and inland sub-markets, across density gradients from dense urban cores to low-density suburbs, and across counties with very different zoning regimes. If ψ varies meaningfully across the city, and if Metro Rail stations happen to be located in areas with systematically different supply conditions, the aggregate welfare estimate could mask important heterogeneity in who captures the gains from transit investment.

Our starting hypothesis was that housing supply conditions are a first-order determinant of the incidence of transit benefits. Near stations with inelastic supply, demand increases are capitalised into land prices and captured by existing landowners; near stations with elastic supply, housing quantities adjust and the benefits are spread more broadly. Understanding which regime prevails matters for both efficiency and distributional assessments of transit policy.

We extend the Severen (2021) replication in two directions. The first is an attempt to estimate ψ separately for geographic sub-regions of the LA metro area, splitting the sample by coastal proximity, 1990 population density quartile, and county group. The second, motivated by the failure of the first, is a systematic sensitivity analysis varying ψ across a wide range in the calibrated GE model.

Our key findings are as follows. The Bartik instrument does not have enough variation within sub-regions to separately identify ψ : first-stage F -statistics fall well below 5 in every geographic partition we try. The paper’s own pooled first stage is moderately strong (Kleibergen–Paap $F = 12.3$ in the preferred specification, though between 7 and 9 in others), so while the pooled estimate is adequately identified, there is little power to spare for sub-sample analysis. Turning to the sensitivity analysis, we find that the aggregate welfare gain from Metro Rail is almost completely insensitive to ψ . Across a $5\times$ range of ψ (0.75 to 4.0), the welfare gain moves by only \$0.33 million — a 0.35 % variation around the \$93.55 million baseline. We show that this insensitivity is a direct consequence of the scale of the intervention and of the closed-city welfare measure: Metro Rail is a small shock in a car-dominated city, and the price adjustments that ψ governs are largely transfers within the city rather than aggregate gains.

We also present “amplified-connectivity” counterfactuals showing that scaling up the estimated

commuting benefit of Metro connectivity raises welfare roughly linearly in the multiplier. At $2\times$ the baseline benefit, aggregate welfare rises by \$105 million per year. This pattern is consistent with network coverage — only 4 % of tracts lie near stations — rather than the magnitude of the per-pair commuting benefit being the binding constraint on Metro Rail’s welfare impact, although we note that confirming this would require a counterfactual that expands coverage directly, which we do not simulate.

The remainder of this paper is organised as follows. Section 2 describes the Severen (2021) framework. Section 3 describes our modifications and methods. Section 4 presents results. Section 5 concludes.

2 The Severen (2021) Framework

2.1 Setting and data

Severen (2021) studies the commuting, labour-market, and housing-market effects of the Los Angeles Metro Rail network, which opened in three phases: the Blue Line (downtown LA to Long Beach) in 1990, and the Green and Red Lines in 1993–2000. The analysis covers 2,552 census tracts in the five-county LA metropolitan area (Los Angeles, Orange, Riverside, San Bernardino, and Ventura counties) over the period 1990–2002.

The main data sources are: (i) the 1990 and 2000 decennial censuses for commuting flows between home and work tracts; (ii) the Longitudinal Employer-Household Dynamics (LEHD) database extended through 2015 for robustness; (iii) the Neighbourhood Change Database (NCDB) for pre-trend tests; and (iv) GIS-computed travel times between all $2,552 \times 2,552$ tract pairs under observed and counterfactual transit networks.

2.2 Commuting flow estimates

The paper estimates a difference-in-differences specification for bilateral commuting flows. The treatment indicator is an indicator for whether both the home tract and the work tract contain (or lie within 250 m or 500 m of) a Metro station. Identification rests on the parallel pre-trends assumption, which is supported by the absence of significant pre-1990 trends in wages or housing values for station-adjacent tracts relative to control tracts.

The paper’s preferred estimates show that flows between tract pairs where both tracts contain a station rose by 11–16 %. Our replication of the gravity specification obtains a both-station effect of 10.0 log points (SE 3.8), stable across eight progressively tighter geographic samples and confirmed by a Poisson pseudo-maximum-likelihood (PPML) specification; adding the proximity ring within 250 m raises the combined estimated effect to approximately 17.6 %.

2.3 Structural elasticities

To translate the reduced-form flow estimates into welfare-relevant parameters, the paper estimates four structural elasticities using a Bartik shift-share IV strategy. The housing supply

parameter ψ is the most consequential. It is estimated by an IV regression of the change in housing prices ($\Delta \ln P_{it}$) on the change in housing density ($\Delta \ln N_{it}$), instrumenting density with a shift-share demand shock based on the 1990 industry mix interacted with national industry-level employment growth. The preferred estimate is $\psi = 1.602$ (SE 0.453). Note that ψ is an *inverse* elasticity — the slope of price on quantity — so the implied housing supply elasticity is $1/\psi = 0.624$: housing supply in LA is inelastic.

The first stage of the preferred specification has a Cragg–Donald F -statistic of 11.5 and a Kleibergen–Paap F -statistic of 12.3; alternative specifications in the paper’s Table 4 range from 5.5 to 12.4. The paper’s bootstrap 95 % confidence interval for ψ is $[1.01, 3.67]$. An internal consistency test — whether ψ estimated from housing consumption and from residential land use are consistent — returns a p -value of 0.616, supporting the structural interpretation.

2.4 General-equilibrium welfare model

The welfare calculation uses the hat-algebra approach of [Dekle et al. \(2008\)](#), extended to a multi-location setting with endogenous commuting. Workers choose a home and work location by drawing Fréchet-distributed idiosyncratic preference shocks, giving rise to a gravity equation for commuting flows:

$$\pi_{ij} \propto \left(\frac{B_i W_j Q_i^{\zeta-1}}{\delta_{ij}} \right)^\varepsilon, \quad (1)$$

where W_j is the wage at workplace tract j , Q_i is the housing price at residence tract i , B_i captures residential amenities, δ_{ij} is the bilateral commuting cost, $1 - \zeta$ is the housing expenditure share, and ε is the Fréchet shape parameter governing the dispersion of idiosyncratic location preferences. Importantly, ε is *estimated* in the paper (at ≈ 2.18) using the same shift-share IV strategy, rather than calibrated from the literature; the labour supply first stage is in fact the weaker of the paper’s IV designs (KP F of 8.0 in the preferred specification). The housing market clears at each tract with inverse supply elasticity ψ . The model is calibrated to the observed 2000 LA equilibrium and asked: by how much does welfare change when the observed transit improvements are removed? The answer — solved by fixed-point iteration converging in 6–11 iterations — is the welfare gain attributable to Metro Rail. The baseline result is **\$93.55 million per year in aggregate** (closed-city, 2016 dollars), matching the \$94 million reported in the paper.¹ This is the figure we take as our benchmark throughout.

3 Modifications and Methods

3.1 Motivation: heterogeneous housing supply

[Severen \(2021\)](#) estimates a single pooled ψ for the five-county LA metro area. The implicit assumption is that housing supply elasticity does not vary systematically with station proximity or with the geographic characteristics of the sub-market in which a station is located. This assump-

¹The dollar figure is the model’s percentage welfare gain (0.044 %) scaled by mean annual earnings (\$31,563) and the 6.73 million workers in the five-county area. It corresponds to roughly \$14 per worker per year.

tion is strong: coastal tracts face stricter development constraints than inland tracts; already-dense central-city tracts have less room for upward densification than low-density suburban tracts; and LA County faces very different zoning conditions than Riverside or San Bernardino counties.

If ψ is systematically *higher* near stations — supply more inelastic, e.g. because stations were placed in already-dense, supply-constrained areas — the paper’s aggregate welfare estimate understates the extent to which transit benefits are capitalised into land values rather than shared broadly. Understanding the distribution of ψ across LA’s sub-markets therefore matters for both efficiency and distributional assessments of the Metro Rail programme.

3.2 Attempted sub-regional estimation

We attempt to estimate ψ separately for three types of geographic sub-regions, using the same Bartik IV specification as the original paper:

- **Coastal vs. inland.** Tracts within 5 km of the Pacific coastline versus all other tracts (296 coastal tracts, $\approx 2,257$ inland). Coastal distances are computed from a 1:50 000 Natural Earth coastline using California Albers projection.
- **Density quartiles.** Tracts are ranked by 1990 homeowners per km² and split at the median (binary) and into quartiles. The prior is that already-dense tracts face less elastic supply (higher ψ) than low-density tracts.
- **County groups.** A *core* group (Los Angeles and Orange counties: dense, heavily zoned) versus a *fringe* group (Riverside, San Bernardino, and Ventura counties: lower density and more developable land). Both groups are large and industrially diverse, so the Bartik instrument was expected to retain adequate within-group variation.

Each specification uses the same IV as the original paper (the leave-one-out Bartik employment shock z_{it}), restricts to the estimation sample ($tc_hval = 0$), and weights by 1990 homeowners.

3.3 Identification failure: instrument weakness

Every sub-regional split encounters the same problem: the Bartik instrument loses power when the sample is partitioned. Table 2 reports the Kleibergen–Paap Wald F -statistic for each split. The paper’s pooled first stage (KP $F = 12.3$ in the preferred specification) is moderately strong, and our inland specification — which retains $\approx 93\%$ of tracts and so nearly proxies the full sample — achieves KP $F = 9.89$. But every genuine partition falls to $F < 5$, and the coastal tracts, Density Q1, and the fringe counties individually fall to $F < 1$.

The root cause is that the Bartik instrument derives its power from cross-industry variation in employment growth interacted with the pre-period industry mix. When the sample is restricted to a geographic sub-area, the effective number of tracts drops substantially, reducing the cross-sectional variation available to identify the supply curve. The coastal split is the extreme case: with only 188 tracts, the instrument is essentially uninformative (KP $F = 0.18$).

This exercise also calibrates how much identifying power the pooled design has to spare: not

much. With KP F statistics between roughly 7 and 12 across the paper’s own specifications and a wide bootstrap confidence interval for ψ ($[1.01, 3.67]$), the pooled estimate is adequately but not overwhelmingly identified — and there is clearly no surplus power for the sub-regional heterogeneity analysis we attempted.

3.4 GE sensitivity analysis

Since sub-regional estimation is infeasible, we vary ψ across a grid spanning 0.5 to 4.0 and re-solve the full Severen GE model at each grid point. Because ψ is an inverse elasticity, this grid spans highly *elastic* supply at the low end ($\psi = 0.5$ implies a supply elasticity of $1/\psi = 2.0$, Sun Belt-type) to very *inelastic* supply at the high end ($\psi = 4.0$ implies $1/\psi = 0.25$, San Francisco-type). All other parameters are held at the paper’s baseline values. At each ψ we verify that:

- the contraction condition for the fixed-point iteration holds (largest eigenvalue of the transition matrix < 1);
- closed-city and open-city equilibria converge (typically in 6–11 iterations to a tolerance of 10^{-5}).

The eigenvalue ranges from 0.778 ($\psi = 0.5$) to 0.951 ($\psi = 4.0$), so the iteration converges stably throughout. (Stability of the iteration is distinct from global uniqueness of the equilibrium; the model’s separate parameter condition for uniqueness is also satisfied at every grid point.)

3.5 “Amplified-connectivity” counterfactuals

As a complementary exercise, we ask how welfare would change if Metro Rail generated a larger commuting benefit than actually observed (e.g. through faster or more reliable service).

The commuting effect enters the model through the pair-specific bilateral utility shifter D_{ij} . In the hat-algebra counterfactual framework, the change in D_{ij} is governed by the estimated commuting parameters λ_{00}^D and λ_{02}^D , recovered from a difference-in-differences regression of log bilateral commuting flows on binary connectivity indicators T_{ij}^{00} and T_{ij}^{02} —equal to one when both home tract i and workplace tract j lie within 500 m of a station of the respective Metro line—controlling for OD-pair, residence-tract-by-year, and workplace-tract-by-year fixed effects.

For multiplier $s \geq 1$, the counterfactual adjusts D_{ij} as:

$$\hat{D}_{ij} = \exp\left(\lambda_{00}^D(s-1)T_{ij}^{00} + \lambda_{02}^D(s-1)T_{ij}^{02}\right), \quad (2)$$

where $\lambda_{00}^D = 0.149$ and $\lambda_{02}^D = 0.128$. Since both parameters are positive, $\hat{D}_{ij} > 1$ for Metro-connected pairs at $s > 1$, increasing their pair-specific commuting utility. At $s = 1$ the expression equals one (no change); at $s = 2$, each connected pair receives $(s-1) = 1$ additional unit of the estimated commuting utility gain—doubling the improvement attributed to Metro. For OD pairs that are not Metro-connected ($T_{ij}^{00} = T_{ij}^{02} = 0$ — over 99.9 % of pairs, carrying ≈ 99.2 % of commuters), $\hat{D}_{ij} = 1$ regardless of s .

It is worth noting that every Metro-connected pair receives the same proportional utility shift, independent of actual trip length or actual travel time, because the estimated λ^D parameters capture a flat average treatment effect. The GE model is re-solved in full at each s , so downstream price, wage, and population adjustments are endogenous. This counterfactual does not literally test trains running at a physically higher speed; it asks what equilibrium would result if Metro connectivity raised commuting utility by s times its estimated magnitude. We accordingly refer to s as an amplification multiplier, retaining “speed” only as informal shorthand.

Starting from the 2000 calibrated baseline, we add $(s - 1)$ additional units of the estimated benefit to all Metro-connected origin–destination pairs. At $s = 2$ the commuting disutility reduction is twice its estimated magnitude. The GE model is re-solved at each multiplier and the incremental welfare gain on top of the \$93.55 million baseline is recorded.

Re-solving the GE model across multipliers $s \in \{1.25, 1.5, 2, 3\}$ yields welfare gains of \$25–\$223 million per year, increasing approximately linearly in s . This pattern is consistent with network coverage—only 94 of 2,552 tracts (4 %) lie near a station—rather than the magnitude of the per-pair commuting benefit being the binding constraint on Metro Rail’s welfare impact. We note, however, that this comparison is indirect: we amplify the benefit on existing corridors but do not simulate an expansion of the network itself, which would be the direct test.

4 Results

4.1 Aggregate welfare is insensitive to ψ

Table 1 reports the closed-city welfare gain from Metro Rail across the ψ grid. The headline finding is that aggregate welfare is essentially invariant to ψ . Moving from $\psi = 0.75$ to $\psi = 4.0$ — a more than $5\times$ increase — reduces the welfare estimate by only \$0.33 million, from \$93.71 million to \$93.38 million. This represents a 0.35 % variation around the paper’s baseline of \$93.55 million. Over the full grid (0.5 to 4.0), the variation is \$0.41 million.

The relationship is monotone: welfare is slightly *higher* under more elastic supply ($\psi = 0.5$, supply elasticity 2.0: \$93.79 million) than under more inelastic supply ($\psi = 4.0$, supply elasticity 0.25: \$93.38 million). The intuition is that with elastic supply, housing quantities near stations can expand, letting more households take up the improved commutes; with inelastic supply, more of the local adjustment is absorbed by price increases, which in the closed-city model are transfers to incumbent landowners rather than aggregate gains. Either way the aggregate effect is tiny. The open-city population gain varies somewhat more (0.102 % at $\psi = 0.5$ down to 0.081 % at $\psi = 4.0$): more elastic supply accommodates more in-migration in response to the same transit improvement.

Table 1: Welfare sensitivity to the inverse housing supply elasticity (ψ)

ψ	Closed-city welfare (\$M/year)	Gain (% of income)	Open-city pop. gain
0.50	\$93.79	0.0442%	0.102%
0.75	\$93.71	0.0441%	0.097%
1.00	\$93.65	0.0441%	0.093%
1.25	\$93.60	0.0441%	0.091%
1.50	\$93.56	0.0440%	0.089%
1.602 (baseline)	\$93.55	0.0440%	0.088%
1.75	\$93.53	0.0440%	0.087%
2.00	\$93.50	0.0440%	0.086%
2.50	\$93.46	0.0440%	0.084%
3.00	\$93.43	0.0440%	0.083%
4.00	\$93.38	0.0440%	0.081%

Notes: GE model re-solved at each ψ value; all other parameters held at [Severen \(2021\)](#) baseline. Baseline row in bold. Welfare is aggregate annual welfare for the five-county area in millions of 2016 dollars. “Gain (% of income)” is the model’s closed-city percentage welfare benefit (compensating variation as a share of income). Lower ψ corresponds to more elastic housing supply. Source:

run_welfare_psi_sensitivity.R.

4.2 Sub-regional ψ estimates: instrument weakness

Table 2 reports the sub-regional ψ point estimates and first-stage F -statistics. The pattern is unambiguous: every geographic partition that reduces the sample substantially pushes the Kleibergen–Paap F -statistic below the conventional threshold of 10, and several produce $F < 1$. The coastal split, with only 188 tracts, is completely uninformative ($F = 0.18$); the ψ point estimate of 1.49 has a standard error of 5.72 and is statistically indistinguishable from any value. Even the county-group splits, designed to preserve within-group industry diversity, produce F -statistics of 3.24 (core) and 1.93 (fringe).

The most nearly reliable sub-regional estimate is the inland specification ($\hat{\psi} = 0.49$, SE 0.17, KP $F = 9.89$), which proxies the full sample because 93 % of tracts are inland. Notably, this estimate is substantially *below* the paper’s pooled $\psi = 1.602$, and the density-quartile specifications where the instrument is marginally adequate (Q2, Q4) give estimates in the range 0.54–0.59, also below 1.602. Recalling that ψ is an inverse elasticity, lower values mean *more elastic* housing supply: if these marginally identified estimates were taken at face value, housing supply in LA would be more elastic than the paper’s pooled estimate implies. Given Section 4.1, however, the aggregate welfare estimate would be essentially unchanged in that case.

Table 2: Sub-regional inverse housing supply elasticity (ψ) estimates

Sub-region	N tracts	$\hat{\psi}$	SE	KP F
Full sample (pooled, Severen)	2,175	1.602	0.453	12.3
Core: LA + Orange Co.	$\approx 1,850$	0.170	0.097	3.24
Fringe: Riverside + SB + Ventura	≈ 590	-0.650	0.489	1.93
Coastal (< 5 km Pacific)	188	1.488	5.716	0.18
Inland (≥ 5 km Pacific)	$\approx 2,254$	0.486	0.169	9.89
High density ($\geq$ median, 1990)	$\approx 1,200$	0.435	0.174	1.83
Low density ($<$ median, 1990)	$\approx 1,200$	0.493	0.400	7.46
Density Q1 (least dense)	≈ 600	0.088	1.080	0.007
Density Q2	≈ 600	0.590	0.311	6.48
Density Q3	≈ 600	0.367	0.240	2.58
Density Q4 (most dense)	≈ 600	0.540	0.255	6.47

Notes: Each sub-regional row estimates ψ by IV (ivreg2, Bartik instrument, weighted by 1990 homeowners). KP F = Kleibergen–Paap Wald F -statistic. Conventional threshold for adequate identification: $F \geq 10$. Negative estimates for fringe counties reflect near-zero first stages and should not be given structural interpretation. Full-sample row reproduces the preferred specification of Severen (2021), Table 4 (column 6). Source: tracts_elasticities_coastal.do, tracts_elasticities_density.do, tracts_elasticities_countygroup.do.

4.3 “Amplified-connectivity” counterfactuals

Table 3 reports the welfare gains from amplifying the Metro Rail commuting benefit, expressed as the incremental gain on top of the \$93.55 million baseline from current transit. The relationship is roughly linear: doubling the reduction in disutility ($s = 2\times$) adds \$105 million per year, approximately four times the gain from a 25 % amplification (\$25 million).

Table 3: Welfare gains from amplified Metro Rail connectivity

Multiplier s	Incremental gain (\$M/year)	Total welfare (\$M/year)	Open-city pop. gain
1.0× (current)	\$0	\$93.55	0.000%
1.25×	\$25.17	\$118.72	0.024%
1.50×	\$51.09	\$144.64	0.048%
2.00×	\$105.26	\$198.81	0.099%
3.00×	\$223.47	\$317.02	0.211%
20.0×	\$5,921.70	\$6,015.25	—

Notes: Incremental gain = additional aggregate welfare on top of current transit (\$93.55 million/year). The multiplier s applies an additional disutility reduction of $(s - 1) \times$ the estimated effect on Metro corridors. Open-city population gains are relative to the current-transit equilibrium. The GE model converges at all levels of s . Source: `simulate_faster_transit.R`; 20× from `map_faster_transit_20x.R`.

The 20× counterfactual is extreme but instructive. Aggregate welfare reaches \$5,922 million per year, roughly 63 times the current transit benefit and about 2.8 % of total annual labour earnings in the metro area. Even at this scale the response remains close to linear in s : within any plausible range of transit improvements, the model’s general-equilibrium feedbacks do not generate non-linear amplification of the commuting benefit.

4.4 The mechanism: why ψ does not move aggregate welfare

The insensitivity of aggregate welfare to ψ follows from two features of the exercise: the small scale of the intervention, and the nature of the closed-city welfare measure.

First, scale. LA Metro Rail induces roughly a 10–18 % increase in commuting flows between station pairs, but those pairs are a small fraction of all $2,552 \times 2,552$ tract-pair combinations — only 4 % of tracts are near a station. The aggregate demand shock to housing near stations is therefore small relative to the total LA housing stock.

Second, what ψ actually governs. Monocentric city simulations (using actual LA tract geography and an Alonso–Muth–Mills demand structure) make the role of ψ transparent. For a local housing demand shock D_n at tract n , with demand elasticity η :

$$\Delta \ln P_n = \frac{\psi D_n}{1 + \eta \psi}, \quad \Delta \ln N_n = \frac{D_n}{1 + \eta \psi}. \quad (3)$$

A higher ψ (more inelastic supply) pushes more of the local adjustment into prices and less into quantities. This split determines the *incidence* of the transit gains — how much is capitalised into land values and captured by incumbent landowners versus accommodated by new housing and shared with movers — but it contributes little to the *aggregate* welfare number. In the closed-city measure, the price component is predominantly a transfer between residents and incumbent

landowners and largely nets out; the aggregate gain is dominated by the direct commuting-cost reduction on connected pairs, which does not involve ψ at all. Only when the demand shock is very large (as in the $20\times$ counterfactual) do the GE price adjustments become large enough for ψ to have a visible aggregate effect.

This points to an important caveat on our own exercise. Showing that aggregate welfare is insensitive to ψ does *not* show that ψ is irrelevant: the distributional question that motivated this study — who captures the gains from transit — is governed precisely by the price-vs-quantity split in Equation (3). Answering it would require decomposing the model’s gains by tract group and by owner/renter status, which the aggregate welfare scalar cannot reveal. We leave that decomposition to future work.

5 Conclusion

This paper replicates the [Severen \(2021\)](#) analysis of LA Metro Rail welfare and extends it by asking whether the city-wide assumption on housing supply elasticity affects the results. Our answer is that it does not affect the aggregate welfare estimate, and we show why.

The attempt to estimate ψ sub-regionally reveals a data limitation: the Bartik instrument, moderately strong at the pooled level (KP $F = 12.3$ in the preferred specification, with a wide bootstrap confidence interval for ψ), loses power entirely when the sample is partitioned by geography or density. This limits what can be learned about heterogeneity in housing supply conditions across LA’s sub-markets from this instrument and this dataset.

The GE sensitivity analysis, however, shows that this limitation is largely inconsequential for the paper’s headline welfare estimate. Across a $5\times$ range of ψ , aggregate welfare moves by \$0.33 million — 0.35 % of the \$93.55 million baseline. The mechanism is transparent: Metro Rail is a small intervention in a large, car-dominated city, and the price adjustments that ψ governs are largely transfers within the closed city rather than aggregate gains. We emphasise that this is a statement about the aggregate: ψ continues to govern the incidence of the gains — the split between capitalisation into land values and broadly shared housing-quantity adjustment — which the aggregate welfare scalar cannot detect.

The amplified-connectivity counterfactuals add a policy implication. Even doubling the reduction in commuting disutility would add only \$105 million per year, and the response is linear in the multiplier up to extreme values (at an imaginary $20\times$ amplification the gain reaches \$5,922 million). The pattern is consistent with network coverage, rather than the per-pair magnitude of the commuting benefit, being the binding constraint on Metro Rail’s welfare impact — though a direct test would require simulating network expansion.

Taken together, these findings reinforce a central message of [Severen \(2021\)](#). The LA Metro Rail’s welfare impact is modest because the city’s commuting patterns are fundamentally shaped by car infrastructure, and a limited rail network serving 4 % of census tracts cannot substantially reshape spatial equilibrium. Housing supply elasticity, however it varies across the metro area,

is not the lever that would change this conclusion.

References

- Bartik, T. J. (1991). *Who Benefits from State and Local Economic Development Policies?* W. E. Upjohn Institute for Employment Research, Kalamazoo, MI.
- Dekle, R., Eaton, J., and Kortum, S. (2008). Global rebalancing with gravity: Measuring the burden of adjustment. *IMF Staff Papers*, 55(3):511–540.
- Diamond, R. and McQuade, T. (2019). Who wants affordable housing in their backyard? An equilibrium analysis of low-income property development. *Journal of Political Economy*, 127(3):1063–1117.
- Goldsmith-Pinkham, P., Sorkin, I., and Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624.
- Monte, F., Redding, S. J., and Rossi-Hansberg, E. (2018). Commuting, migration, and local employment elasticities. *American Economic Review*, 108(12):3855–3890.
- Severen, C. (2021). Commuting, labor, and housing market effects of mass transportation: Welfare and identification. *Review of Economics and Statistics*, forthcoming. https://doi.org/10.1162/rest_a_01037.